

IoT-Based Weed Detection & Removal Robot

Professional Project Execution Guide

1. Project Overview

Build a 4WD autonomous weed detection and removal robot. Phase 4 focuses on robot control and automation. Phase 5 focuses on AI-based weed detection using YOLO and wireless camera streaming.

2. Current Repository Structure

```
docs/  
hardware/ESP-32_sketch_apr12a/sketch_apr12a.ino  
hardware/components-list.md  
software/backend/server.js  
software/backend/routes/  
software/backend/data/logs.json  
software/frontend/index.html  
software/frontend/pages/*  
software/frontend/js/*  
software/frontend/css/*
```

3. Final Architecture

Phase 4:
Dashboard → Backend → ESP32 → L298N → Motors
Dashboard → Backend → ESP32 → Relay → Cutter

Phase 5:
LILYGO T-SIMCAM OV5640 → Wi-Fi → HP Victus RTX3050 → YOLOv8 → ESP32 → Robot

4. Hardware Bill of Materials (BOM)

ESP32 DevKit V1
LILYGO T-SIMCAM ESP32-S3 + OV5640 (Phase 5)
2× L298N Motor Drivers
4WD Chassis Kit
4 DC Gear Motors
RS775 Cutter Motor
Relay Module
2× HC-SR04 Sensors
12V Battery Pack
LM2596 Buck Converter
Emergency Stop Switch
Wires, Connectors, Mounts

5. Estimated Budget

Phase 4: ₹6,000–7,000
Phase 5 Vision Upgrade: ₹3,000–4,000
Expected Total Project Cost: ₹9,000–11,000

6. Team Structure

Member 1 – Backend Developer
Member 2 – Frontend Developer
Member 3 – ESP32 Developer
Member 4 – QA, Documentation & Integration

7. Detailed File Ownership

Member 1:
software/backend/*

Member 2:
software/frontend/*

Member 3:
hardware/ESP-32_sketch_apr12a/*
hardware/components-list.md

Member 4:
docs/*
Testing Reports
Project Documentation

8. Backend Tasks

Create movement APIs.
Create cutter APIs.
Robot status APIs.
Logging APIs.
API documentation.
Integration support.

9. Frontend Tasks

Control dashboard.
Movement controls.
Cutter controls.
Status monitoring.
Logs page.
Responsive UI.

10. ESP32 Tasks

Motor control.
L298N integration.
Relay integration.
HC-SR04 integration.
Obstacle avoidance.
Wi-Fi communication.
Status updates.

11. QA & Documentation Tasks

Prepare wiring diagrams.
Test plans.
Bug tracking.
Integration testing.
Demo preparation.

Final report preparation.

12. Shared Hardware Assembly

All members participate in chassis assembly, motor mounting, wiring, power system installation, cutter installation and testing.

13. Robot Movement Logic

Forward, Backward, Left, Right, Stop.
Differential drive architecture using two L298N modules.
Zero-radius turning supported.

14. API Contract

POST /api/forward
POST /api/backward
POST /api/left
POST /api/right
POST /api/stop
POST /api/cutter/on
POST /api/cutter/off

15. Git Workflow

main = stable branch
develop = integration branch
backend-dev
frontend-dev
esp32-dev
docs-dev

16. Software Installation Checklist

Git
VS Code
Node.js
Arduino IDE
ESP32 Board Package
Postman
Python 3.11 (Phase 5)
OpenCV (Phase 5)
YOLOv8 (Phase 5)
PyTorch (Phase 5)

17. Week-by-Week Plan

Week 1: Hardware procurement and setup.
Week 2: Backend, frontend, ESP32 development.
Week 3: Integration and cutter control.
Week 4: Testing and delivery.

18. Testing Strategy

Movement testing.
Cutter testing.

Obstacle detection testing.
API testing.
Integration testing.
Stress testing.

19. Phase 5 Roadmap

Install LILYGO T-SIMCAM.
Create dataset.
Train YOLOv8.
Real-time weed detection.
ESP32 command automation.
Field testing.

20. Risks & Mitigation

Power issues → Validate battery system.
Wi-Fi disconnects → Local hotspot network.
Hardware failure → Keep spare wires and connectors.
YOLO accuracy → Create custom dataset.

21. Final Success Criteria

Robot moves correctly.
Cutter works safely.
Obstacle detection works.
Dashboard controls robot.
Logs are generated.
YOLO detects weeds (Phase 5).
System demonstration completed successfully.